9.3 Colouring Fence Posts

WORKSHEET

The question

STEP II 1995, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q2

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Pause and try Count the ways to paint 3 posts in a circle, and 4. Does $3 \times 2^{n-1}$ overcount, and by how much?

STAGE 1 Two kinds of colouring

Covers: why $r_{n+1} = s_n$, and finding $r_n + s_n$

Pause and try Take a colouring of $n+1$ posts with the first and last posts red. What colour can post n be? Remove the last post: which kind of colouring of n posts is left? Can you go back the other way?

Ask yourself $r_n + s_n$ counts every colouring of n posts with the first post red. How many choices are there for each post after the first?

STAGE 2 The formula by induction

Covers: the induction

Ask yourself Check $n = 1$. Then use $r_{n+1} = 2^{n-1} - r_n$ with the formula for r_n, and simplify.

STAGE 3 The circle

Covers: posts in a circle

Pause and try Fix post 1 red and cut the circle between post n and post 1, so the posts are in a line. Which extra condition must the line colouring satisfy? Which of r_n and s_n counts those? How many colours can post 1 have?

What's really going on

Chromatic polynomials

The posts and their adjacencies form a *graph*: a path for the line, a cycle for the circle. A *proper colouring* gives adjacent vertices different colours. The number of proper colourings with k colours is a polynomial in k, the *chromatic polynomial* of the graph, introduced by George Birkhoff in 1912. Repeating the question's argument with k colours gives

$$\text{path: } k(k-1)^{n-1}, \qquad \text{cycle: } (k-1)^n + (-1)^n(k-1).$$

With $k = 3$ the cycle formula is $2^n + 2(-1)^n$. Birkhoff hoped to prove the four colour theorem (every map can be coloured with four colours so that neighbouring countries differ) by showing that the chromatic polynomial of every map's graph is positive at $k = 4$. That approach never succeeded; the theorem was finally proved in 1976 by Kenneth Appel and Wolfgang Haken, with extensive computer help. Chromatic polynomials remain an active field, with surprising connections to statistical physics, where they appear as the *Potts model* of magnetism.

The same count by matrices

There is another way to see the answer, using linear algebra from Chapter 7. Let $\mathbf{M}$ be the 3×3 matrix with 0 on the diagonal and 1 elsewhere: entry (i, j) is 1 when colour j may follow colour i. Then entry (i, j) of $\mathbf{M}^{n-1}$ counts line colourings that start with colour i and end with colour j, so r_n is a diagonal entry of $\mathbf{M}^{n-1}$. A circle colouring is a sequence that returns to its start after n steps, so the number of circle colourings is the *trace* of $\mathbf{M}^n$. Since $\mathbf{M} = \mathbf{J} - \mathbf{I}$, where every entry of $\mathbf{J}$ is 1, the eigenvalues of $\mathbf{M}$ are $3 - 1 = 2$ (eigenvector $(1, 1, 1)$) and -1 (twice), and the trace of $\mathbf{M}^n$ is the sum of the

nth powers of the eigenvalues:

$$\operatorname{tr} \mathbf{M}^n = 2^n + (-1)^n + (-1)^n = 2^n + 2(-1)^n.$$

This *transfer matrix* method is used throughout combinatorics and statistical physics to count arrangements built up one step at a time.

Try this next

1. **k colours.** Repeat the question with k colours. Show that $r_{n+1} = s_n$ still holds and that $r_{n+1} + r_n = (k-1)^{n-1}$, find r_n, and deduce the chromatic polynomial of the cycle, $(k-1)^n + (-1)^n(k-1)$.

2. **No two reds together.** How many ways are there to paint n posts in a line red or white, with no two adjacent posts red? (You should find the Fibonacci numbers.)

3. **A wheel.** Add a central post adjacent to every post of the circle (a "wheel"). How many ways are there to colour it with 4 colours? (*Hint:* the centre uses one colour, and the circle must use the other three.)

4. **Necklaces.** If two colourings of the circle count as the same when one is a rotation of the other, how many different colourings are there for $n = 4$? For $n = 5$? (This is the start of *Burnside's lemma*.)

Further reading

- **Robin Wilson, *Introduction to Graph Theory*** (Pearson). The standard first course; covers colourings and the four colour problem.
- **Robin Wilson, *Four Colours Suffice*** (Penguin). The story of the four colour theorem, for a general reader.
- **Herbert Wilf, *generatingfunctionology*** (free online). Recurrences and how to solve them, including the transfer matrix method.

The full guided solution is in Section 9.3:
https://beyondstep.keithmcnulty.org/ch09/fence-posts.html